

AS5: Scrape hierarchical pages

Part 4:

- Scrape several features from two types/levels of pages
- Challenging level: ★★★★★

Two layers of pages

ONE QUOTE PER DAY.

[Home](#) [Random](#) [Archive](#)

So long...

...and thanks for all the fish.

Today's quote was the last one - after 12 years (and some change), I am closing this blog down. As in, there won't be any new posts, but you can still hit the ["Random"](#) button each day to read one (or more) of the 4395 quotes collected on this blog. There's definitely a bunch of duplicates but it's the thought that counts.



January 1, 2023 / 15 notes

"A book of quotations ... can never be complete."

—Robert M. Hamilton

[#Hamilton](#) [#quote](#) [#quotes](#)

January 1, 2023 / 10 notes

"Cheers to a new year and another chance for us to get it right."

—Oprah Winfrey

[#Winfrey](#) [#quote](#) [#quotes](#)

December 31, 2022 / 7 notes

Menu page

ONE QUOTE PER DAY.

[Home](#) [Random](#) [Archive](#)

"A book of quotations ... can never be complete."

—Robert M. Hamilton

[#Hamilton](#) [#quote](#) [#quotes](#)

January 1, 2023 / 10 notes

Detail page

- fantasyworldin liked this
- the-hope-of-a-new-day reblogged this from onequoteperday
- hola-signorared16 liked this
- caffeine-anxiety-and-randomfacts liked this
- samsaintjames reblogged this from onequoteperday
- whatreality liked this
- onequoteperday posted this

Leer theme by [Renz Abong](#)

Collect data across two levels of pages

ONE QUOTE PER DAY.

Home Random Archive

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January 1, 2023 / 15 notes

"A book of quotations ... can never be complete."

—Robert M. Hamilton

#Hamilton #quote #quotes

January 1, 2023 / 10 notes

Menu page

ONE QUOTE PER DAY.

Home Random Archive

"A book of quotations ... can never be complete."

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#Hamilton #quote #quotes

January 1, 2023 / 10 notes

Detail page

 fantasyworldin liked this

 the-hope-of-a-new-day reblogged this from onequoteperday

 hola-signorared16 liked this

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 onequoteperday posted this

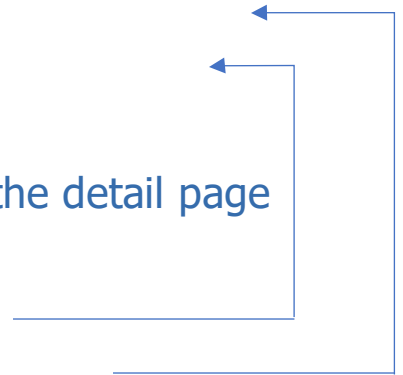
Quote	Author
"A book of quotations ... can never be complete."	Robert M. Hamilton
"Nobody can be exactly like me. Sometimes even I have trouble doing it."	Tallulah Bankhead
"We have just enough religion to make us hate, but not enough to make us love one another."	Jonathan Swift

Compare with the previous

- An iteration

- Scrape the page
 - Scrape a wrapper
 - Scrape features
 - Next wrapper
 - Click 'older'
- 

- An iteration

- Scrape the page
 - Scrape a wrapper
 - Scrape features
 - Scrape features from the detail page of the wrapper
 - Next wrapper
 - Click 'older'
- 

At wrapper level

ONE QUOTE PER DAY.

Home Random Archive

So long...

...and thanks for all the fish.

Today's quote was the last one - after 12 years (and some change), I am closing this blog down. As in there won't be any new posts, but you can still hit the "Random" button and play to see (one or more) of the 135 quotes collected on this blog. There's definitely a bunch of duplicates but it's the thought that counts.



January 1, 2023 / 15 notes

"A book of quotations ... can never be complete."

—Robert M. Hamilton

#Hamilton #quote #quotes

January 1, 2023 / 10 notes

```
><div class="post text">...</div>
▼<div class="post quote">
  <span>"A book of quotations ... can never be complete."</span>
  <div class="source">—Robert M. Hamilton</div>
  ><div class="tags">...</div>
  ▼<div class="post_footer">
    <a href="https://onequoteperday.tumblr.com/post/705235970434416640/a-book-of-quotations-c
an-never-be-complete">January 1, 2023</a> == $0
    " / "
    <a href="https://onequoteperday.tumblr.com/post/705235970434416640/a-book-of-quotations-c
an-never-be-complete#notes">10 notes</a>
  </div>
  <!-- Post Footer -->
</div>
```

```
def parse(self, response):
    wrappers = response.xpath('//div[@class="post quote"]')
    for wrapper in wrappers:
        quote = wrapper.xpath('span/text()').extract_first()
        author = wrapper.xpath('div[@class="source"]/text()').extract_first()
        lower_url = wrapper.xpath('div[@class="post_footer"]/a/@href').extract_first()
        yield Request(lower_url, callback= parse_lower)

        yield {'Quote': quote, 'Author': author}

    next_rel_url = response.xpath ('//div[@id="pagination"]/center/a/@href').extract()[-1]
    next_url = response.urljoin(next_rel_url)
    yield Request(next_url, callback=self.parse)
```

```
def parse_lower(self, response):
```

- Define a new function to scrape the detail pages
- All detail pages are alike

At detailed page

"A book of quotations ... can never be complete."

—Robert M. Hamilton

#Hamilton #quote #quotes

January 1, 2023 / 10 notes

fantasyworldin liked this

the-hope-of-a-new-day reblogged this from onequoteperday

hola-signorared16 liked this

caffeine-anxiety-and-randomfacts liked this

samsaintjames reblogged this from onequoteperday

whatreality liked this

onequoteperday posted this

```
<li class="note like tumblelog_fantasyworldin without_comme
  <a rel="nofollow" class="avatar_frame" target="_blank" href="https://www.tumblr.com/fantasyworldin" title="FantasyWorld" ></a>
```

```
<span class="action">
```

```
<a rel="nofollow" href="https://fantasyworldin.tumblr.com/" title="FantasyWorld">
  fantasyworldin</a> == $0
```

```
" liked this "
```

```
</span>
```

```
<div class="clear"></div>
```

```
</li>
```

```
<li class="note reblog tumblelog_the-hope-of-a-new-day without_commentary"></li>
```

```
<li class="note like tumblelog_hola-signorared16 without_commentary"></li>
```

```
<li class="note like tumblelog_caffeine-anxiety-and-randomfacts without_commentary"></li>
```

```
def parse(self, response):
    wrappers = response.xpath('//div[@class="post quote"]')
    for wrapper in wrappers:
        quote = wrapper.xpath('span/text()').extract_first()
        author = wrapper.xpath('div[@class="source"]/text()').extract_first()
```

```
lower_url = wrapper.xpath('div[@class="post_footer"]/a/@href').extract_first()
```

```
yield Request(lower_url, callback= self.parse_lower,
               meta={'Quote': quote, 'Author': author})
```

```
yield {'Quote': quote, 'Author': author}
```

```
next_rel_url = response.xpath ('//div[@id="pagination"]/center/a/@href').extract()[-1]
```

```
next_url = response.urljoin(next_rel_url)
```

```
yield Request(next_url, callback=self.parse)
```

```
def parse_lower(self, response):
    notes = response.xpath('//span[@class="action"]/a/text()').extract()
    response.meta['Notes'] = notes
    yield response.meta
```

- meta parameter: a dictionary used to pass data through requests
- That data is then accessible inside the callback function via response.meta

MGP AS4

- Visit <https://quotes.toscrape.com/>
- Create the five spider in this `quote` project to scrape all the quotes, authors, and each quote's first tag on **all the menu pages**, together with the authors' birthday and bio description on **all the detail pages** linked by the [\(about\)](#) after the authors' name on the menu page
- Save it in a CSV file

